

JEOPARDY! WAGERING STRATEGY

A plain-English guide to smarter Daily Double and Final Jeopardy bets

Built for 3-player games, with 2-player and 1-player play as reduced cases

Not affiliated with Jeopardy! or Sony Pictures Television • A study and practice tool, not a live-assist device

Two moments decide the game

Everything else is trivia and buzzer speed. These two moments are where a dollar amount, not a fact, decides who wins.

$ Daily Double

Found mid-game, while a category is in play. You choose the wager before you know the exact clue.

The decision affects the rest of the game, not just the next 30 seconds -- more clues, maybe another Daily Double, then Final Jeopardy after that.

? Final Jeopardy

The last clue of the game. Every player wagers, then answers the same category.

A single terminal event -- the math here is fully determined, with a clean guarantee-based answer available before you even see the clue.

Two different questions, not one

This project treats them as two separate tools. Blending them into one answer hides which one you actually asked for.

SAFE BET

“What wager can't hurt me if I'm right?”

A guarantee against the worst realistic case. Needs no estimate of your odds of being correct.

SMART BET

“Given how sure I really am, what wins most often?”

A probability-weighted decision. Uses your confidence and a model of how opponents tend to play.

Safe Bet: the guarantee

Assume every rival makes their single most threatening legal play: bet everything, answer correctly.

You	Rival 1	Rival 2
$12,000	**$8,000**	**$6,000**

Rival 1's worst case: $8,000 doubled = $16,000. That's the biggest threat -- bigger than Rival 2's $12,000 -- so it's the one you must cover.

SAFE BET RECOMMENDS

$4,001

$12,000 + $4,001 = $16,001 -- one dollar past the worst case

Smart Bet: using your confidence

Same scores. Smart Bet also counts a tie for first as a win -- matching the real show rule that co-champions both advance.

AT 50% CONFIDENCE

$4,000

~78% estimated win rate

recommended wager stays the same -- confidence changes how good the bet is, not how much to bet here

AT 90% CONFIDENCE

$4,000

~96% estimated win rate

recommended wager stays the same -- confidence changes how good the bet is, not how much to bet here

A rule worth knowing: never settle for a tie when a win is free.

If two wagers reach the same win probability, but one only ties an opponent while the other beats them outright at no extra cost, Smart Bet now always prefers the outright win.

Daily Double: a different kind of bet

A Final Jeopardy wager affects the next 30 seconds. A Daily Double wager affects everything still to come.



This wager → More clues in play → Maybe another Daily Double → Final Jeopardy

There's no clean formula for that many moving parts. So the engine simulates the rest of the game forward, thousands of times, for every candidate wager -- and recommends whichever wins most often.

The next slide shows why that simulation has to be read carefully.

A 12x swing that shouldn't happen

Same scores. Confidence moved two points. The old engine's recommendation should not have moved this much.

42% confidence

$350

44% confidence

$4,200

BEFORE THE FIX -- a real bug, not a hypothetical

AFTER THE FIX -- both confidence levels

$0

The engine now tracks each estimate's statistical uncertainty, and only trusts a difference it can actually detect.

Even long shots get modeled correctly

You're in 3rd place. Can you still win Final Jeopardy if both leaders bet big and miss?

You	Rival 1	Rival 2
$4,000	**$9,000**	**$10,000**

At 100% self-confidence, the model finds you win exactly when both rivals bet big and get it wrong -- fully independent events.

1 in 12

~8.3% chance to win from 3rd place

matches the exact number the app showed -- verified by hand, not just by the code's own output

Try it yourself

One file, opens in any browser, no install and no internet needed: app/index.html

Daily Double

Enter scores, board money, and confidence -- get a wager and a win-probability curve.

Final Jeopardy

Compare Safe Bet and Smart Bet side by side for any 1-, 2-, or 3-player game.

Quiz Me

Random scenarios, graded A+ to D against the optimal play, with streaks.

How This Works

A plain-language explanation of every tier, linked back to the full docs.

What's proven, what's a placeholder

Told plainly, on purpose -- the goal is a tool you can trust precisely as much as it's earned.

SOLID

- Safe Bet's guarantee math -- fully determined, no confidence input needed
- Smart Bet's exact win-probability math, for a given opponent model
- The Daily Double engine's statistical selection -- no longer fooled by simulation noise
- 30 automated tests, hand-derived where possible -- not just checked against the code's own output

PLACEHOLDER

- How opponents actually tend to wager and answer -- a labeled guess, not fitted to real data yet
- Regular-clue and future-Daily-Double dynamics -- simplified, ignores buzzer competition
- The JavaScript app's parity with the Python engine -- checked by hand, not by automated CI yet

Get started

1. Open app/index.html in any browser
2. Pick Daily Double or Final Jeopardy
3. Enter scores and your confidence
4. Get a recommendation, or drill it in Quiz Me



Want the full math?

Every claim on these slides is proven or cited in the white paper -- worked examples, full derivations, and references.

docs/white-paper.md

Also see: docs/tutorial/, docs/how-to/, docs/reference/, and docs/explanation/ for task-by-task guides